

Fake News Detector with Model Interpretability

Introduction

The rapid growth of misinformation across social media platforms causes significant threat to the public and societal trust. An automated fake news detection offers a data driven approach to flagging out misinformation before it spreads. This project explores different interpretable classification models and tries to figure out the best strategies to use in building a fake news detector with an emphasis on understanding why a model reaches that conclusion.

Objective

The primary objective is to design, train, and evaluate multiple interpretable machine learning models on a recent dataset of real-world news articles. The specific goals include:

- Identify and preprocess a well-labeled, fake news dataset that reflects current patterns of misinformation.
- Apply regularization for principled feature selection, reducing dimensionality while preserving predictive signal.
- Train and compare Logistic Regression, SVM (with kernel methods), and at least one additional classifier, reporting performance across standard metrics.
- Integrate a RAG-based explanation module that retrieves relevant context and generates human-readable rationales for each prediction.
- Produce a clear, reproducible codebase and a written report documenting methodology, results, and findings.

Proposed Tasks

Keeping goal of the project in mind and the based on the topics covered during lectures, the tasks for this project are as follows:

Dataset Finding and Cleaning	Handling missing values, performing text normalization, and more.
Feature selection using regularization techniques	Extracting Linguistic features, applying LASSO regularization
Tuning Hyperparameters	Using Cross validation to tune regularization strength and kernel parameters
Model Training and Evaluation	Train models, visualize performance, report accuracy
Incorporating RAG for explaining prediction	Build a retrieval-augmented generation pipeline that provides supporting evidence for prediction
Documentation and Report	Write a project report covering motivation, methods, results and discussion

Timeline

The project spans five weeks with each week corresponding to a focused deliverable:

Week 1 - Dataset Finding and Cleaning

Week 2 - Feature selection using regularization techniques and Tuning Hyperparameters

Week 3 - Model Training and Evaluation

Week 4 - Incorporating RAG for explaining prediction

Week 5 - Documentation and Report

Summary

This project addresses a real-world problem of detection of fake news by the application of machine learning techniques covered in this course. By leveraging feature selection regularization, cross-validation for model selection, and kernel-based SVMs alongside logistic regression, the project will deliver comparative study of interpretable classifiers. The integration of a RAG-based explanation system sets this project apart from a normal classifier by adding explainability to the predictions made.

References

"Liar, Liar Pants on Fire": A new benchmark dataset for fake news detection.

<https://aclanthology.org/P17-2067>

"Recognizing Fake News":

<https://www.kaggle.com/code/ruchi798/how-do-you-recognize-fake-news/notebook>

"Retrieval-Augmented Generation (RAG) from basics to advanced":

<https://medium.com/@tejpal.abhyuday/retrieval-augmented-generation-rag-from-basics-to-advanced-a2b068fd576c>